the points was chosen, in this example, just for convenience).

The *reachability plot* shown in figure 12.17 is a special plot composed of two interconnected charts: the top one simply shows the final (flat) clustering that we obtain by setting the threshold to ε, while the bottom one, showing the ordered sequence of reachability distances, explains how the clustering was derived; clusters and reachability distances are filled with matching colors, so that it's easier to see what points in the top half match the sections in the bottom half (outliers are marked using black, and for the sake of convenience, to properly plot all values, we assign ε_{MAX} instead of `undefined` to the reachability distance of outliers): by doing so we are, in practice, loosening the requirements for the reachability criterion for these points but, since it's already the largest possible value that we can assign to the threshold ε, it won't change anything in the following steps (described below).

So, how do we form these clusters, given $\varepsilon \leq \varepsilon_{MAX}$? Figure 12.18 illustrates the idea behind this algorithm (at a high level, combining it with elements of OPTICS); we start looking at reachability distances, following the ordering in which points were processed by OPTICS, so that we know that the reachability distance of the next point is the smallest possible among all undiscovered points (keep in mind that the reachability distanced stored are points-to-cluster distances, and that's also why ordering is important!).

We start from the first point processed, call it `P1`, and create a new cluster `C1`; being the first point of a new cluster, `P1`'s